Sine rule

1

a No **b** Yes **c** Yes **d** No

2

a $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$ **b** $\dfrac{p}{\sin P} = \dfrac{q}{\sin Q} = \dfrac{r}{\sin R}$

3

a $\sin A = \dfrac{x}{b}$ **b** $x = a \sin B$ **c** $\dfrac{\sin A}{a} = \dfrac{\sin B}{b}$

Unknown sides

4

a $a = 10.50$ **b** $a = 6.85$ **c** $a = 3.02$ m **d** $a = 711.73$ cm

e $a = 11.29$ cm

5

a $x = 13.1$ **b** $x = 12.9$

6

a

i $a = 16.66$ **ii** $a = 16.66$

b

i $a = 17.14$ **ii** $a = 17.14$

c

i $a = 30.09$ **ii** $a = 30.09$

7

a $HK = 733.17$ km **b** $KJ = 444.55$ km

8

a $\angle A = 75.78$ **b** $b = 3.152$ m **c** $c = 5.723$ m

Unknown angles

9

a $x = 26.0^\circ$ **b** $x = 32.7^\circ$ **c** $x = 50.7$ **d** $x = 30.4^\circ$

e $x = 20.4^\circ$ **f** $x = 30.3^\circ$ **g** $x = 43.2^\circ$ **h** $x = 149.1^\circ$

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i $x = 133.13^\circ$ **j** $x = 23.5^\circ$

10

a $B = 74^\circ$ **b** $B = 31^\circ$

11

a $x = 19.54^\circ$ **b** 48.46°

12

a $x = 41$ **b** 139

13 $\theta = 33.84^\circ$

Applications

14

a $\angle OBA = 39^\circ$ **b** $k = 21.76$

15 $x = 53.6 \text{ m}$

16

a $\angle A = 38^\circ$ **b** $c = 22.71 \text{ m}$ **c** $b = 22.71 \text{ m}$ **d** 1.9 min

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